Siva Sankar Udaya Kumar

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AI & ML Engineer with a strong software engineering background, experienced in building **production scale AI systems, data pipelines and analytics solutions**. Specialized in **GenAI, financial & predictive analytics and decision support**, with hands-on experience delivering **secure, observable services** using **Azure, AWS and modern MLOps practices**.

Experience

Freelance Projects — Dec 2024 – Present

GenAI Credit Risk Agent — *Remote*

- Built a production grade ***GenAI agent*** using ***ASP.NET Core*** and ***Azure OpenAI*** for financial document and credit risk analysis.
- Implemented ***Retrieval Augmented Generation (RAG)*** with ***Azure AI Search*** and ***MCP style agent orchestration*** to ensure grounded, explainable outputs.
- Developed ***secure REST APIs*** with role based access control, structured logging, audit trails and observability to support safe LLM usage in regulated environments.

Real-Time Stock Analytics Pipeline — *Remote*

- Developed a real-time data pipeline using ***Kafka, Airflow, MinIO***, and ***PostgreSQL*** to ingest, store, and process live stock market data from the Finnhub API.
- Designed ***Airflow DAG***s to compute volatility metrics (log returns, Sharpe ratio) and trigger alerts for anomalies such as price drops or abnormal volatility.
- Built a ***Streamlit*** dashboard for real-time monitoring with interactive visualizations of stock trends and alert events.

Norwegian University of Science and Technology — Oct 2021 – Nov 2024

PhD Candidate — *Trondheim, Norway*

- Designed and implemented ***stochastic optimization models*** in ***Python*** to support operational decision making under uncertainty in ***large scale energy systems***.
- Developed ***scalable scenario generation pipelines*** using probabilistic modeling, ***GAN and Reinforcement Learning***, delivering reproducible and performance optimized components for decision support workflows.

Indian Institute of Technology, Bombay — Jul 2021 – Sep 2021

AI Researcher — *Mumbai, India*

- Developed an ***end-to-end vehicle insurance cross-sell prediction system*** using ***Databricks, Apache Spark***, and ***XGBoost***, with MLflow for experiment tracking and model performance optimization.
- Designed and deployed a ***scalable REST API*** for model inference, implementing ***MLOps practices*** with ***Docker*** and ***Kubernetes*** for automated deployment, monitoring, and scalability.

Relatas — Jan 2021 – June 2021

Data Scientist — *Bangalore, India*

- Developed a ***Smart Opportunity Recommender*** tool using ***Python*** and ***scikit-learn***, incorporating ***sentiment analysis*** and ***NLP*** to identify high potential sales leads.
- Deployed models on ***AWS*** for scalability and performance, managing workflows with ***Jira*** in an ***agile*** environment.

Iha Consulting Services — Oct 2020 – Dec 2020

ML Engineer — *Secunderabad, India*

- Developed a ***Human Activity Recognition model*** using Ensemble Learning on multi sensor time series data, performing feature engineering and exploratory analysis to enhance predictive accuracy.
- Built an ***end-to-end MLOps pipeline*** for model training, versioning, deployment, and monitoring, integrating ***CI/CD workflows*** for production grade reliability.

iSmile Technologies — June 2020 – Sep 2020

AI Engineer — *Bolingbrook, United States (Remote)*

- Led a ***team of three*** in integrating Computer Vision and Robotics for autonomous object detection using a custom model built with ***Azure Custom Vision***.
- Developed an ***Android app*** for real-time detection and coordinated delivery using ***Azure DevOps*** and ***SAFe***, ensuring streamlined collaboration and project execution.

CodeSpeedy Technology — Jan 2020 – June 2020

AI Developer — *Kolkata, India (Remote)*

- Built and deployed ***end-to-end ML systems*** across ***NLP, Computer Vision and Time Series*** using ***Python, TensorFlow*** and ***AWS/Azure***.
- Operationalized models with ***CI/CD, observability*** and ***scalable API based serving*** in production environments.

Education

Norwegian University of Science and Technology Oct 2021 (ABD)
Ph.D. in Industrial Economics and Technology Management *Trondheim, Norway*
- Thesis: Short-Term Optimization under Uncertainty in the Norwegian Natural Gas System

Indian Institute of Technology (BHU), Varanasi Jul 2019 – June 2021
M.Tech. in Industrial Management (GPA: 9.11) *Varanasi, India*
- Thesis: Automated Detection and Tracking of Sewer Pipe Problems Using Inspection Videos

Jain University Jul 2015 – June 2019
B.Tech. in Aerospace Engineering (GPA: 8.01) *Bangalore, India*

Technical Skills

Programming Languages: Python, SQL, C# (.NET), MATLAB, Julia
GenAI & ML Systems: Azure OpenAI, LLMs, RAG, Agent Orchestration (MCP), REST API
Cloud & Infra: Databricks, Azure Cloud, GCP, AWS, Docker, Kubernetes, Terraform
Big Data & Streaming: Apache Spark, Kafka, Airflow, PostgreSQL, MinIO, Hive, BigQuery
MLOps & Observability: MLflow, CI/CD, Git, GitLab, Jira
ML & AI Frameworks: TensorFlow, Scikit-learn, PyTorch, XGBoost
Dashboarding & Spreadsheet: PowerBI, Streamlit, Data Studio, Tableau, MS Excel
Spoken Languages: Tamil, English, Norwegian, Hindi, Telugu, Kannada